

Make everything super nice number (e.g. bin widths, chi bounds, L, etc. are integers)

Hopefully this way Mathematica will have an easier time evaluating e.g. eigenvalues which it couldn't do before from the mess that comes out when you have special integrals Si/Ci/Ei with fractional/irrational arguments

While I choose specific numbers here, the eigenvalue structure, more specifically the number of zero-eigenvalues, should be general

```
In[ ]:= (*i=1,...,n*)
chimin = 1;
chimax = 4;
L := chimax - chimin;
nbins = 3;
dchi := L / nbins;
l[i_] := chimin + (i - 1) dchi
r[i_] := chimin + i dchi
fLo[m_, α_] := 1 / L Sin[Pi / L α (1 + 2 m)] / Sin[Pi / L α]

In[ ]:= assume = chimin ∈ PositiveReals && chimax ∈ PositiveReals &&
chimin < chimax && chi ∈ PositiveReals && chip ∈ PositiveReals && chimin < chi &&
chimin < chip && chi < chimax && chip < chimax && li ∈ PositiveReals &&
ri ∈ PositiveReals && li < ri && chimin < li && chimin < ri && li < chimax &&
ri < chimax && lj ∈ PositiveReals && rj ∈ PositiveReals && lj < rj &&
chimin < lj && chimin < rj && lj < chimax && rj < chimax && i ∈ PositiveIntegers &&
chimin + i dchi ≤ chimax && j ∈ PositiveIntegers && chimin + j dchi ≤ chimax;

In[ ]:= unfiltered = DiagonalMatrix[Table[1 / dchi ^ 2 (1 / l[i] - 1 / r[i]), {i, 1, nbins}]]

Out[ ]:= {{1/2, 0, 0}, {0, 1/6, 0}, {0, 0, 1/12}}
```

Only remove the mean

```
In[ ]:= $Assumptions = assume;
fmatm0[i_, j_] := KroneckerDelta[i, j] (1/l[i] - 1/r[i]) 1/dchi^2 +
  1/L^2 (1/chimin - 1/chimax) - 1/(L dchi) (1/l[i] - 1/r[i] + 1/l[j] - 1/r[j])
covmatm0 = Table[fmatm0[i, j], {i, 1, nbins}, {j, 1, nbins}] // FullSimplify;

(*also could use rank-nullity*)
MatrixRank[covmatm0] == nbins - 1
```

Out[]:= True

```
In[ ]:= Eigenvalues[covmatm0]
```

Out[]:= $\left\{ \frac{1}{36} (9 + \sqrt{21}), \frac{1}{36} (9 - \sqrt{21}), 0 \right\}$

```
In[ ]:= $Assumptions = assume;
{u, σ, v} = SingularValueDecomposition[covmatm0];
σ // FullSimplify // MatrixForm
```

Out[]:= MatrixForm=

$$\begin{pmatrix} \frac{1}{36} (9 + \sqrt{21}) & 0 & 0 \\ 0 & \frac{1}{36} (9 - \sqrt{21}) & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

Setup functions for m > 0

```
In[ ]:= $Assumptions = assume;
fCrossUnbinned[m_, chi_, chip_] :=
  Evaluate[FullSimplify[fLo[m, chi - chip] (1/chi^2 + 1/chip^2)]]

fLoLoIntegrand[m_, chi_, chip_] :=
  Evaluate[FullSimplify[1/chib^2 (fLo[m, (chi - chib)] * fLo[m, (chip - chib)])]]

(*consistency check with m=0*)
Integrate[fLoLoIntegrand[0, chi, chip], {chib, chimin, chimax}] ==
  1/L^2 (1/chimin - 1/chimax) // FullSimplify
```

Out[]:= True

If filter out mean and fundamental

```

In[ ]:= m1crossElem[i_, j_] := m1crossElem[i, j] = m1crossElem[j, i] =
      Integrate[fCrossUnbinned[1, chi, chip], {chi, l[i], r[i]}, {chip, l[j], r[j]}]

In[ ]:= m1cross = Table[m1crossElem[i, j], {i, 1, nbins}, {j, 1, nbins}];

In[ ]:= $Assumptions = assume;
      m1LoLoUnbinned[chi_, chip_] :=
      Evaluate[Integrate[fLoLoIntegrand[1, chi, chip], {chib, chimin, chimax}]]

In[ ]:= m1LoLoElem[i_, j_] := m1LoLoElem[i, j] =
      m1LoLoElem[j, i] = Integrate[m1LoLoUnbinned[chi, chip], {chi, l[i], r[i]}, {chip, l[j], r[j]}]

In[ ]:= m1LoLo = Table[m1LoLoElem[i, j], {i, 1, nbins}, {j, 1, nbins}];

In[ ]:= covmatm1 = unfiltered + m1LoLo - m1cross;

In[ ]:= Eigenvalues[covmatm1] // N

Out[ ]:= {0.109585, 0.058408, 0.}

```

If filter out mean and kF and 2kF

```

In[ ]:= m2crossElem[i_, j_] := m2crossElem[i, j] = m2crossElem[j, i] =
      Integrate[fCrossUnbinned[2, chi, chip], {chi, l[i], r[i]}, {chip, l[j], r[j]}]

In[ ]:= m2cross = Table[m2crossElem[i, j], {i, 1, nbins}, {j, 1, nbins}];

In[ ]:= $Assumptions = assume;
      m2LoLoUnbinned[chi_, chip_] :=
      Evaluate[Integrate[fLoLoIntegrand[2, chi, chip], {chib, chimin, chimax}]]

In[ ]:= m2LoLoElem[i_, j_] := m2LoLoElem[i, j] =
      m2LoLoElem[j, i] = Integrate[m2LoLoUnbinned[chi, chip], {chi, l[i], r[i]}, {chip, l[j], r[j]}]

In[ ]:= m2LoLo = Table[m2LoLoElem[i, j], {i, 1, nbins}, {j, 1, nbins}];

In[ ]:= covmatm2 = unfiltered + m2LoLo - m2cross;

In[ ]:= Eigenvalues[covmatm2] // N

Out[ ]:= {0.0568136, 0.0228974, 0.}

```